American International University-Bangladesh (AIUB)

Department of Computer Science

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Computer Graphics [A]

Final-Term Project Report

PROJECT TITLE:

Dreamland Village Ultimate

An Interactive 2D Animated Village Simulation Using OpenGL & C++

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Group: 07 **Section:** A

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1 Introduction

1.1 Project Overview

Dreamland Village Ultimate is an interactive 2D animated village made with C++, OpenGL and FreeGLUT. It shows a village of Bangladesh with houses, trees, a paddy field, animals, a river and a boat. No images or textures are used. Everything is drawn from basic shapes: points, lines, triangles, quads and polygons.

The project has four scene modes: Day, Sunset, Night and Rain. The user can switch between them at any time with the keyboard. Each mode changes the sky colours, the sun or moon, the effects (stars, fireflies, rain, lightning) and the background sound, while the village animation keeps running.

1.2 Objectives

- Practise drawing basic shapes, circles and ellipses, and using transformations (translate, rotate, scale).
- Draw a Bangladeshi village with cottages, banyan, palm and banana trees, a paddy field and a river.
- Use simple math (sine waves) to animate the cow, birds, trees, water, smoke, clothes and the boat.
- Make a rain effect with 260 falling drops, splashes on the ground and random lightning.
- Keep the animation smooth with a timer that updates the scene about 60 times per second.

1.3 Tools & Technologies Used

Component	Used for
Programming Language	C++ (C++11)
Graphics	OpenGL for drawing, GLU for the 2D projection (gluOrtho2D)
Window & Events	FreeGLUT 3.0.0 (MinGW package, included in the repository): window, keyboard, mouse, timer and text
Sound	Windows PlaySound (winmm) to play .wav files
Math	C++ <cmath> (sin, cos, fabs, fmod)
Compiler & IDE	MinGW GCC with Code::Blocks on Windows 64-bit
Version Control	Git and GitHub

2 Project Proposal

2.1 Problem Statement

A village scene has many things moving at the same time: light changes during the day, water waves, trees moving in the wind and animals walking. Drawing all of this in pure 2D without any images needs math for the shapes and motion, the right drawing order, and an animation loop that stays smooth.

2.2 Proposed Solution

We split the project into five parts: basic shapes and sound, landscape, animals and details, river and weather, and main control. Each part is a set of drawing functions. The display function calls them from back to front, and a timer function updates all the moving values.

3 Coordinate System & Drawing Order

3.1 Coordinate System Setup

The program uses a 2D orthographic projection: `gluOrtho2D(0, 100, 0, 80)`. So the world is 100 units wide and 80 units high, with (0, 0) at the bottom-left corner. The window is 1280 x 720 pixels, and OpenGL scales the world to fit the window.

3.2 Drawing Order (Back to Front)

There is no depth buffer, so objects that are drawn later appear in front (painter's algorithm). The display() function draws in this order:

1. **Sky:** sky gradient, sun or moon, sun rays, stars, clouds and birds (Y about 38 to 80).
2. **Land:** hills, paddy field, chicken, grass and the cow (Y about 17 to 52).
3. **Village:** path, palm trees, both houses, chimney and smoke, fences, banyan tree, banana plants, flower garden and clothesline (Y about 17 to 62).
4. **River:** river with waves, ripples, sun/moon reflection, fish and the bridge (Y about 0 to 20).
5. **Boat & ducks:** rowing boat with the fisherman, and ducks (Y about 5 to 16).
6. **Effects on top:** fireflies, rain, lightning, rain splashes and the help panel.

4 List of Objects

ID	Object	OpenGL Primitives	Description
OBJ-01	Sky	GL_QUADS	Two-colour gradient that changes with the scene mode.
OBJ-02	Sun & Rays	GL_POLYGON, GL_LINES	Yellow sun circle with 12 rays that slowly rotate.
OBJ-03	Moon & Stars	GL_POLYGON, GL_POINTS	Crescent made from two circles; stars as points and small twinkling circles.
OBJ-04	Clouds	GL_POLYGON (ellipses), GL_QUADS	White (grey in rain) ellipse groups moving slowly to the right.
OBJ-05	Birds	GL_LINE_STRIP	V-shaped birds flying across the sky with flapping wings.
OBJ-06	Hills & Field	GL_TRIANGLES, GL_POLYGON, GL_LINES	Hills, green land, yellow-green paddy field and swaying grass.
OBJ-07	Cottages	GL_QUADS, GL_POLYGON, GL_LINES	Two houses with roofs, doors and windows; windows glow at night.
OBJ-08	Trees	GL_LINE_STRIP, GL_POLYGON	Banyan, palm and banana trees whose leaves sway in the wind.
OBJ-09	Cow	GL_POLYGON, GL_TRIANGLES, GL_QUADS, GL_LINES	Cow walking left and right with moving legs, nodding head and swinging tail.
OBJ-10	River	GL_QUAD_STRIP, GL_LINES, GL_LINE_STRIP	Blue river with a moving sine-wave top edge and ripples.
OBJ-11	Boat	GL_POLYGON, GL_LINES	Wooden boat with a fisherman rowing an oar; can be steered.
OBJ-12	Rain	GL_LINES, GL_LINE_STRIP	260 slanted rain drops, ground splashes and random lightning.

5 Functions to Represent the Objects

5.1 Part 1: Basic Shapes, Sound & Intro (Iftekhar Rahman)

- `rect(x1, y1, x2, y2)`: draws a filled rectangle with GL_QUADS (walls, legs, posts).
- `circle(cx, cy, r, parts)`: draws a filled circle with GL_POLYGON using 70 points by default.
- `ellipse(cx, cy, rx, ry, parts)`: same as circle but with different X and Y radius.
- `line(x1, y1, x2, y2, width)`: draws a line with the given width.
- `drawText(x, y, text, font)`: writes text with glutBitmapCharacter.
- **Sound functions:** `updateSound()` plays nature.wav (Day/Sunset), night.wav (Night) or rain.wav (Rain); `cowSoundTimer()` plays cow.wav every 10 seconds in Day and Sunset; `stopSound()` stops the sound.
- `drawIntro()`: start screen with the project title, a short subtitle, "PRESS ENTER TO START" and the list of modes.

5.2 Part 2: Village Landscape (Najiat Islam Rishad)

- `drawSky(), drawSunMoonStars(), drawSunRays(), drawTwinklingStars()`: sky colours, sun, moon, rotating sun rays and twinkling stars.
- `drawCloud(), drawClouds(), drawBird(), drawBirds()`: moving clouds and flying birds.
- `drawHills(), drawPaddyField(), drawGrassDetails(), drawVillagePath()`: hills, paddy field, grass and the village path.
- `drawPalmTree(), drawBanyanTree(), drawBananaPlant()`: trees with leaves swaying in the wind.
- `drawHouseOne(), drawHouseTwo(), drawChimney(), drawSmoke()`: two cottages, the chimney and rising smoke.

5.3 Part 3: Animals & Village Details (Faysal Talukder)

- `drawCow()`: cow that walks left and right; its legs move, the head nods and the tail swings.
- `drawChicken()`: chicken walking slowly across the yard with a small up-down step.
- `drawFence(), drawFlower(), drawFlowerGarden(), drawClothesline()`: fences, swaying flowers and clothes moving on the line.

5.4 Part 4: River, Boat & Weather (Md. Sazzadur Rahman)

- `drawRiver(), drawRiverRipples(), drawReflection(), drawBridge()`: river with waves, ripples, sun or moon reflection and the bridge.
- `drawBoatBody(), drawFisherman(), drawBoat()`: boat with a fisherman rowing; the boat can face left or right.
- `drawDuck(), drawDucks(), drawOneFish(), drawFish()`: ducks on the water; fish swim across the river and jump out of the water in Rain mode.
- `drawFireflies(), drawRain(), drawRainSplashes()`: fireflies at night, falling rain with lightning, and rain splashes.

5.5 Part 5: Main Control & Animation (Md. Arafat Rahman Samir)

- `display()`: draws the whole scene from back to front.
- `update()`: timer function that runs every 16 ms and changes all the moving values.

- `toggleFullScreen()`: switches between fullscreen and a 1280 x 720 window.
- `keyboard(), specialKeys(), mouse()`: handle keys, arrow keys and mouse clicks.
- `init(), main()`: create the window, set the projection and start the main loop.

6 Algorithms & Math Used

6.1 Drawing Circles and Ellipses

A circle or ellipse is drawn as a polygon with N points (N = 70). Point number i is at angle θi = 2πi / N:

$$xi = cx + rx \cdot cos(\theta i), \quad yi = cy + ry \cdot sin(\theta i), \quad i = 0, 1, \ldots, N-1$$

When rx = ry it is a circle. Different values give ellipses, used for clouds, the cow body, fish and ducks.

6.2 Motion with Sine Waves

Most movement uses a sine wave. A "phase" value grows a little every frame, and the object position or angle follows it:

$$y(t) = y0 + A \cdot sin(phase)$$

- **River waves:** the top edge of the river is y = 16 + 0.65 · sin(0.2x + riverPhase).
- **Cow legs:** the legs move by 0.28 · sin(cowLegPhase), so the front and back legs move in opposite directions.
- **Tree sway:** palm leaves rotate by 2.8 · sin(palmPhase + 0.1x) degrees, so each tree moves a little differently.

6.3 Rotation Around a Point

To rotate something around its own point (pivot), we move the pivot to the origin, rotate, and move it back, using glPushMatrix / glPopMatrix so other objects are not affected:

$$P' = T(px, py) \cdot R(\theta) \cdot T(-px, -py) \cdot P$$

This is used for the sun rays, the swaying trees, the cow head and the fisherman's oar.

7 Controls & Animation Loop

7.1 Keyboard and Mouse

Input	What happens
Enter	Closes the intro screen and starts the village.
1 or D	Day mode: blue sky, sun with rotating rays, nature.wav.
2	Sunset mode: purple-orange sky with a low red sun, nature.wav.
3 or N	Night mode: dark sky, moon, stars, fireflies, night.wav.
4 or R	Rain mode: grey sky, rain, lightning, jumping fish, rain.wav.
B	Starts or stops the boat.
L	Turns the house lights on or off (seen at night).
H	Shows or hides the help panel.
F	Switches fullscreen on or off.
M / U	Turns the sound off / on.

Input	What happens
Left / Right Arrow	Boat moves to the left / right.
Mouse Left / Right click	Boat moves to the right / left.
Mouse Middle click	Stops the boat.
Esc	Stops the sound and closes the program.

7.2 Animation Loop

The animation is driven by a GLUT timer. `update()` runs about every 16 ms (around 60 times per second). It moves the clouds, birds, boat, cow and other objects, updates the rain drops, sometimes starts a lightning flash, then calls `glutPostRedisplay()` and schedules itself again with `glutTimerFunc(16, update, 0)`.

8 Scenes & Output Screenshots

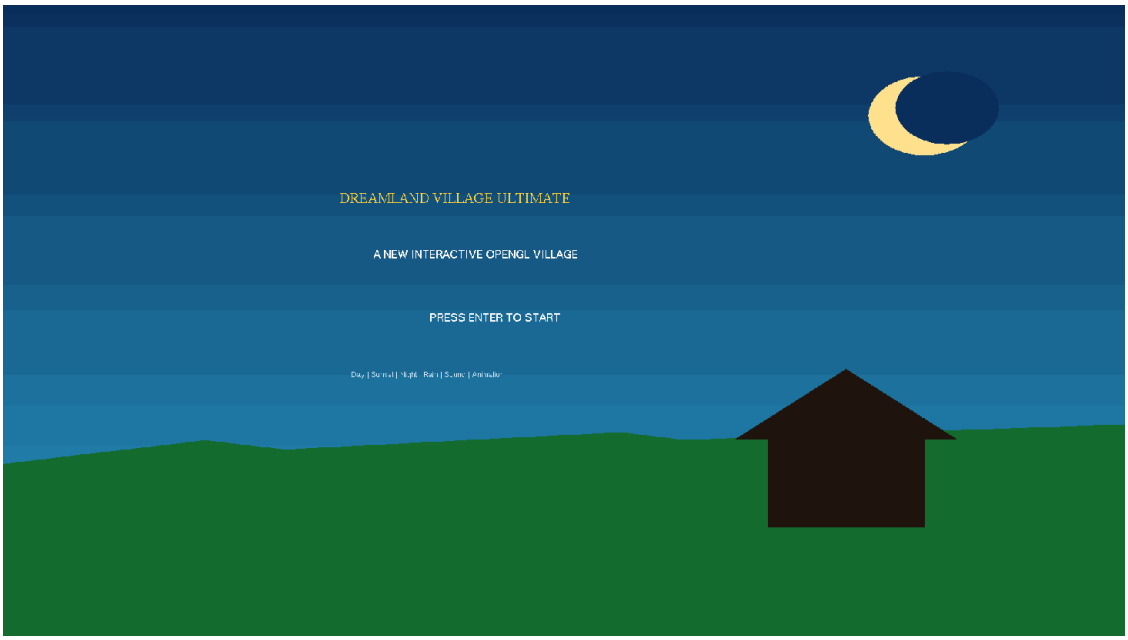


Figure 1: Intro screen with the title and "PRESS ENTER TO START".



Figure 2: Day scene: blue sky, sun with rays, paddy field, cottages, walking cow and the boat.



Figure 3: Sunset scene: purple-orange sky and a red sun going down.



Figure 4: Night scene: dark sky, stars, crescent moon, fireflies and lit house windows.



Figure 5: Rain scene: grey sky, slanted rain, splashes and jumping fish.

9 Limitations & Future Improvements

9.1 Current Limitations

- The boat and animals follow fixed paths; there is no collision between objects.
- All code is in one file (main.cpp) instead of separate classes or files.
- Sound uses the Windows PlaySound function, so it works only on Windows.
- The world is a fixed 100 x 80 area, so in fullscreen on a different screen shape the picture is stretched.

9.2 Future Improvements

- Use a cross-platform sound library such as SDL2 or OpenAL.

- Move the scene to 3D with a camera and lighting.
- Let villagers and animals move around more freely.
- Add more seasons, for example kash flowers in autumn and morning fog in winter.

10 Conclusion

Dreamland Village Ultimate shows the main ideas of 2D computer graphics: drawing basic shapes, circles and ellipses, using transformations, painter's algorithm and timer-based animation. The village runs smoothly at about 60 frames per second and changes between Day, Sunset, Night and Rain with matching sounds.

11 Source Code, Repository & How to Run

Source code, FreeGLUT files, sounds and screenshots:

https://github.com/munshi-rishad/ComputerGraphics-Dreamland-Village-Ultimate

1. Install Code::Blocks with the MinGW compiler (the "mingw-setup" installer).
2. Open `openglportable-64bit/openglportable/openglportable.cbp` in Code::Blocks.
3. Click Build and Run (F9). Press Enter on the intro screen.

The ready-made program is also in `openglportable/bin/Debug/openglportable.exe`; keep the .wav files and freeglut.dll in the same folder.